

Faculty of Science and Technology

PRT563 – Advanced Data Management

Assessment 4: Group Project Report (40%)

Migration of a Relational Database to Neo4j Graph Database

Darwin Multi-Store Supermarket Retail Database

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1. Introduction

This report builds upon the Group Project Plan (Assessment 1) and conceptual modelling and implementation into SQLite (Assessment 2) of the Darwin multi-store supermarket retail project. The initial scope had been on the operational challenges in retailing at supermarkets in the Northern Territory, including long interstate haulage and wet season disruption to the Stuart Highway, the absence of a dedicated local grocery distribution centre, and the need for improved multi-store inventory visibility and supplier performance insight (Yahoo News Australia, 2026; NT Department of Infrastructure, Planning and Logistics, 2023).

In Assessment 2 the team created an Enhanced Entity-Relationship (EER) model, a corresponding UML class diagram, a fully normalised (3NF) relational schema using SQLite (the standard SQL implementation), some sample data, and four SQL use cases, of increasing difficulty. This is the current evaluation of that relational design, moving it onto a Neo4j property graph. The migration will enhance the ergonomics of multi-hop analytical queries and enable the use of algorithms and functions from the new Graph Data Science (GDS) category in centrality and similarity analyses that are hard to code in pure SQL.

The report has the following organization. Section 2 describes the graph data model and the design decisions that were used to translate from the EER model. There is a description of the implementation in Section 3, including constraints, indexes and the pure-Cypher load path. Section 4 introduces four Use Cases in Cypher of increasing complexity. Sections 5 and 6 show different centrality and similarity algorithms from the Neo4j Graph Data Science library, which are compared systematically and then managed and explained. The limitations and future work are discussed in section 7 and the conclusion followed in section 8.

2. Graph Database Modelling

2.1 Translation Approach

The translation followed the standard "rules for converting property graphs to translations" (Robinson, Webber & Eifrem, 2015; Neo4j, n.d.) from property-graphs to translations: Entity types are mapped to node labels, attributes are mapped to node/relationship properties, relationships (including those created from M:N associative entities) are mapped to directed/undirected edges. Additional labels (Cashier, StockClerk, Manager for Employee, LoyaltyMember for Customer) were added to the different specialisations of the EER model, rather than as separate subtype nodes.

2.2 Node Labels and Key Properties

The principal node labels and their identifying properties are:

- **ProductCategory** – categoryId, categoryName
- **Product** – productId, productName, barcode, unitPrice
- **Store** – storeId, storeName, suburb, address, phone
- **Supplier** – supplierId, supplierName, contactName, phone, email
- **Employee** (with additional labels Cashier / StockClerk / Manager)
- **Customer** (with additional label LoyaltyMember) – pointsBalance, tierLevel
- **PurchaseOrder** – purchaseOrderId, orderDate, status
- **SalesTransaction** – saleId, saleDateTime, paymentMethod

2.3 Relationship Types

Key relationships include HAS_PARENT (category hierarchy), IN_CATEGORY, EMPLOYS, PLACES_ORDER, SUPPLIED_BY, CONTAINS (order lines with quantityOrdered and unitCost), HOLDS (inventory with quantityOnHand, reorderLevel and lastRestockDate), PROCESSED, MADE_AT, PURCHASED_BY, INCLUDES (sale lines with quantitySold and unitPriceAtSale), and CO_PURCHASED (materialised weighted co-purchase edges for GDS).

2.4 Design Decisions and Justifications

The graph design has three intentional optimisations:

Inventory as an interrelatedness. The following fields are naturally part of the edge between Store and Product: quantity on hand, reorder level and last restock date. Modelling inventory as a HOLDS relationship will reduce the number of nodes and will make a low-stock path query a single hop.

Additional labelling as specialisations. Extra labels are used to indicate Employee subtypes and LoyaltyMember. Label filtering doesn't involve traversing subtype relationships and continues to be efficient.

Weighted line-item relationships. Based on Neo4j best practice on weighted edges, both CONTAINS and INCLUDES relationships have quantity and price.

Sales data was converted into CO_PURCHASED relationships taking weight as number of shared baskets, to run centrality analysis efficiently.

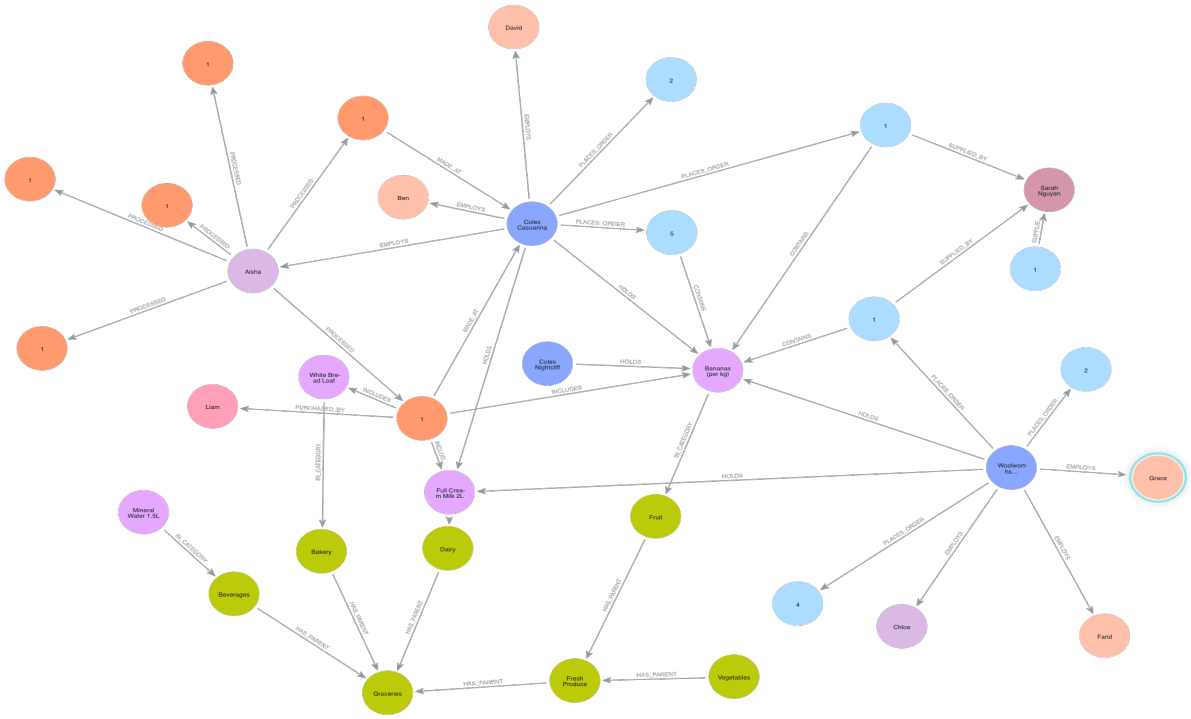


Figure 1 Neo4j Bloom visualisation of the Darwin multi-store supermarket property graph (Danala Group 17)

3. Graph Database Implementation

The relational sample data was moved to Neo4j Desktop (Neo4j 5.x) by loading it into Neo4j Desktop using a pure-Cypher CREATE / MATCH / MERGE load path for maximum portability. Unique constraints were added to all natural keys, and secondary indexes were added to productName, suburb, status, tierLevel and saleDateTime. 86 nodes and 206 relationships were found after loading and verified.:

Table 1 Node counts after migration.

Label	Count
Product	22
SalesTransaction	18
ProductCategory	10
PurchaseOrder	8
Customer / LoyaltyMember	7 + 3
Supplier	5
Employee (all roles)	10
Store	3
Total nodes	**86**

Table 2 Relationship counts after migration.

Relationship Type	Count
INCLUDES	42
CO_PURCHASED	28
IN_CATEGORY	22
PROCESSED / MADE_AT	18 each
PURCHASED_BY	17
HOLDS	15
CONTAINS	11
EMPLOYS	10
HAS_PARENT	9
PLACES_ORDER / SUPPLIED_BY	8 each
Total relationships	**206**

All the import scripts are included in the accompanying repository with the original CSV files, called import.txt for reproducibility.

4. Graph Database Queries

Four Cypher use cases of increasing complexity were executed against the loaded graph.

4.1 Use Case 1 – Simple (Promotion candidates)

Business scenario: Identify fruit and vegetable products priced under $5.00 for a mid-week promotion.

Result: Six products were returned (Table 3). These items form a practical shortlist for a fresh-produce promotion during constrained interstate supply.

Table 3 Use Case 1 results.

Product	Price ($)	Category
Avocado (each)	2.40	Fruit
Carrots 1kg	2.60	Vegetables
Broccoli (each)	3.10	Vegetables
Potatoes 2kg	3.80	Vegetables
Bananas (per kg)	3.90	Fruit
Apples Pink Lady 1kg	4.90	Fruit

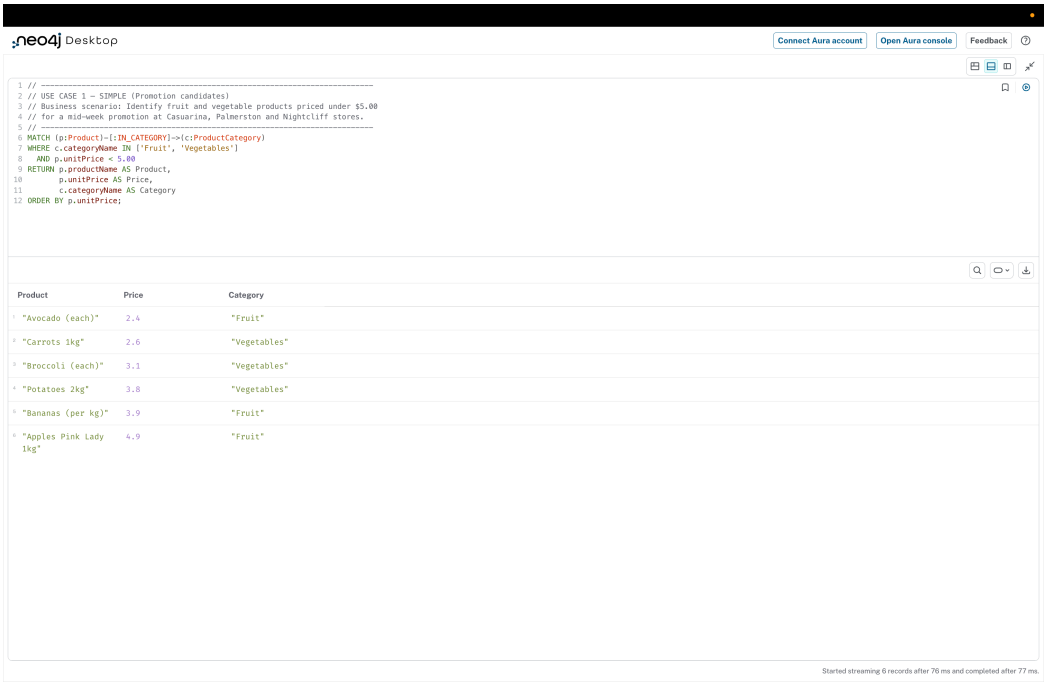


Figure 2 Use Case 1 results – Fruit and Vegetables under $5.00

4.2 Use Case 2 – Simple / Label filter (Loyalty engagement)

Business scenario: Identify Bronze and Silver loyalty members with fewer than 1,000 points for a points-boost campaign.

Result: Five members were returned (Table 4). Gold members were correctly excluded.

Table 4 Use Case 2 results.

Name	Tier	Points	ID
Sophia Martin	Bronze	180	6
Mia Chen	Bronze	420	5
Ava Williams	Silver	650	4
Lucas Ahmed	Silver	780	9
Noah Patel	Silver	890	3

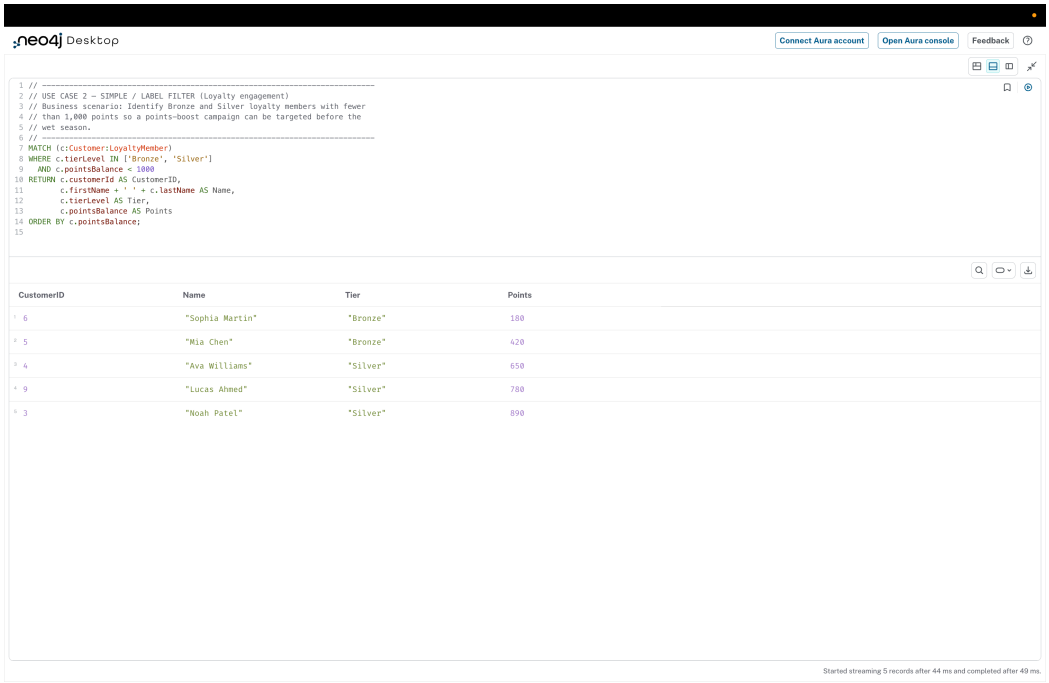


Figure 3 Use Case 2 results – Bronze/Silver loyalty members under 1000 points

4.3 Use Case 3 – Moderately complex (Low stock + recent demand)

Business scenario: List products below reorder level at any store and show units sold in the last seven days, supporting emergency inter-store transfers when the Stuart Highway is disrupted.

Result: Six low-stock lines were identified (Table 5). UnitsSoldLast7Days returned zero because sample dates fall outside the rolling window; the query still correctly identifies stock-at-risk items.

Table 5 Use Case 3 results.

Store	Product	OnHand	Reorder	Sold
Coles Casuarina	Mangoes (per kg)	6	12	0
Coles Casuarina	Avocado (each)	8	15	0
Coles Casuarina	Laundry Powder 2kg	8	10	0
Coles Nightcliff	Bananas (per kg)	9	15	0
Coles Casuarina	Bananas (per kg)	12	25	0
Woolworths Palmerston	Bananas (per kg)	18	20	0

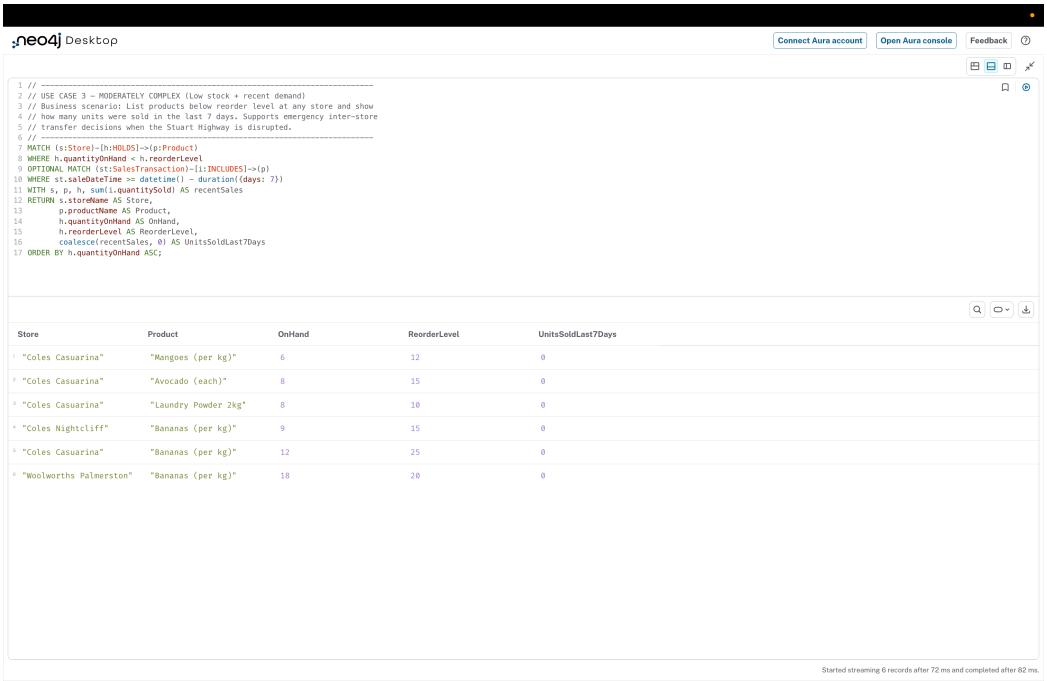


Figure 4 Use Case 3 results – Low-stock products

4.4 Use Case 4 – Complex (Supplier prioritisation)

Business scenario: Rank suppliers by total ordered value of products currently below reorder level.

Result: NT Fresh Distributors ranked highest for Bananas ($1,575), followed by Tropical Fruit Traders ($1,107) (Table 6).

Table 6 Use Case 4 results.

Supplier	Product	Value ($)
NT Fresh Distributors	Bananas (per kg)	1,575.00
Tropical Fruit Traders	Bananas (per kg)	1,107.00
Aussie Household Essentials	Laundry Powder 2kg	340.00
NT Fresh Distributors	Mangoes (per kg)	256.00
Tropical Fruit Traders	Mangoes (per kg)	217.00
NT Fresh Distributors	Avocado (each)	150.00

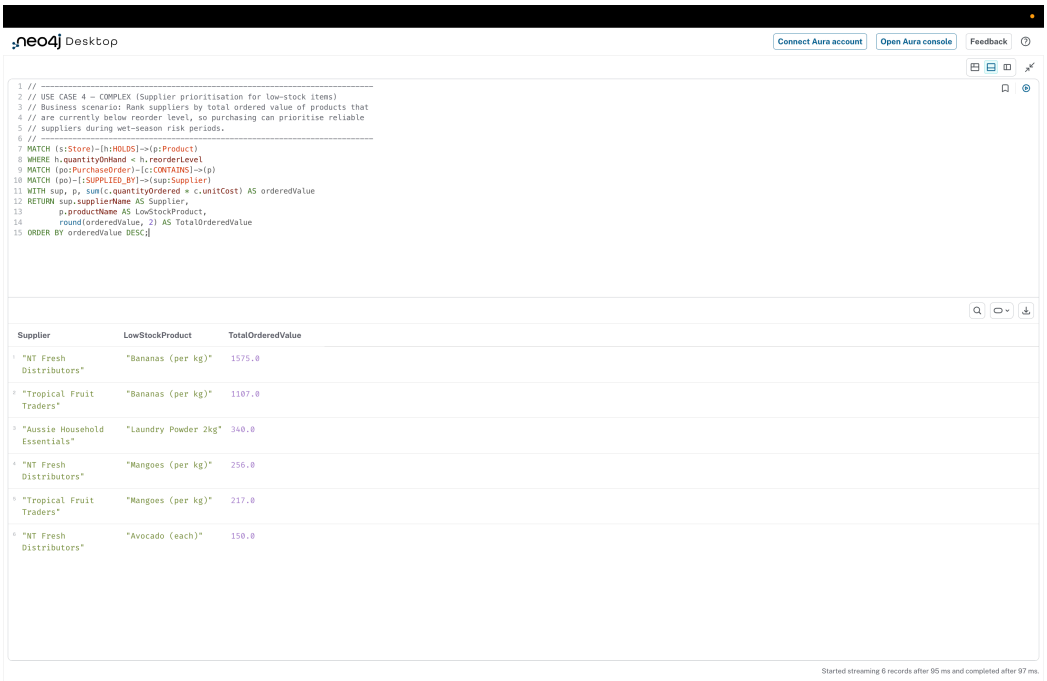


Figure 5 Use Case 4 results – Suppliers of low-stock products by ordered value

5. Centrality Algorithms (GDS)

Business question: Which products act as hubs in customer baskets? High-centrality products should be prioritised for continuous availability during wet-season disruptions.

Method: CO_PURCHASED relationships were materialised (weight = shared baskets). A weighted undirected projection (coPurchase) was created. Three algorithms were run: weighted PageRank, weighted Degree Centrality, and Betweenness Centrality.

5.1 PageRank (weighted)

Top 10 results:

Table 7 Weighted PageRank results.

Product	PageRank
Bananas (per kg)	2.6273
Full Cream Milk 2L	1.9961
Mineral Water 1.5L	1.3422
Apples Pink Lady 1kg	0.9721
Free Range Eggs 12pk	0.9297
Toilet Paper 12pk	0.8995
Carrots 1kg	0.7463
Avocado (each)	0.6892
Orange Juice 2L	0.6755
Broccoli (each)	0.6746

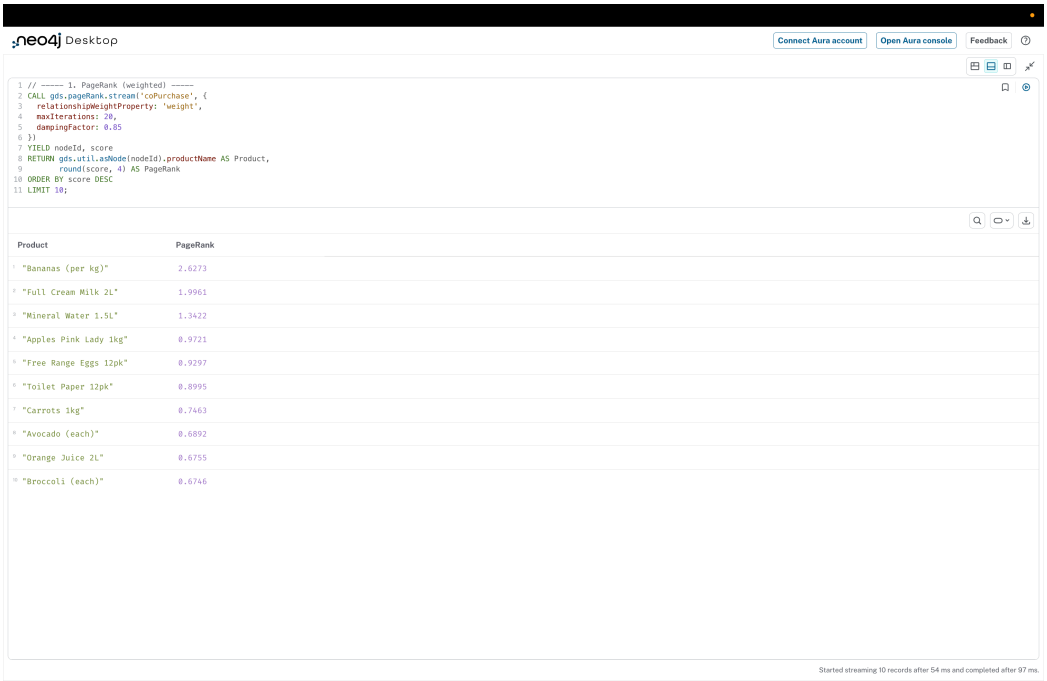


Figure 6 PageRank centrality results

5.2 Degree Centrality (weighted)

Top 10 results:

Table 8 Weighted Degree Centrality results.

Product	Weighted Degree
Bananas (per kg)	18.0
Full Cream Milk 2L	14.0
Mineral Water 1.5L	8.0
Apples Pink Lady 1kg	6.0
Free Range Eggs 12pk	6.0
Toilet Paper 12pk	6.0
Avocado (each)	4.0
Carrots 1kg	4.0
Broccoli (each)	4.0

Orange Juice 2L	4.0

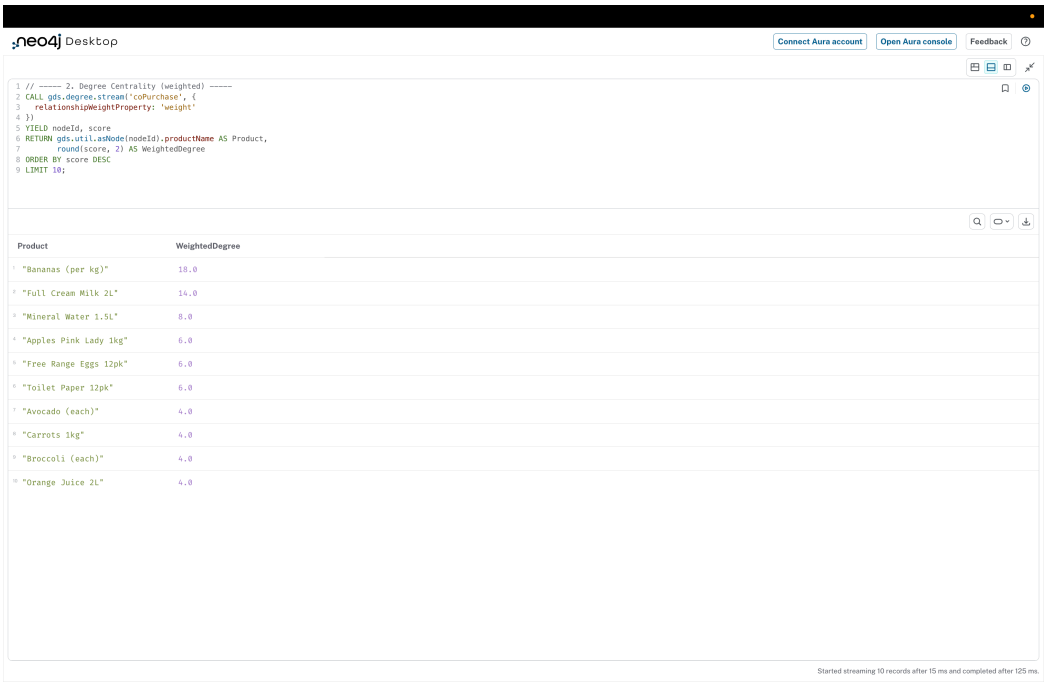


Figure 7 Degree Centrality results

5.3 Betweenness Centrality

Top 10 results:

Table 9 Betweenness Centrality results.

Product	Betweenness
Bananas (per kg)	48.8333
Full Cream Milk 2L	8.6667
Mineral Water 1.5L	8.1667
Carrots 1kg	4.3333
Free Range Eggs 12pk	4.1667
Orange Juice 2L	1.6667
Apples Pink Lady 1kg	1.0000
Broccoli (each)	0.8333
Toilet Paper 12pk	0.3333
Avocado (each)	0.0000

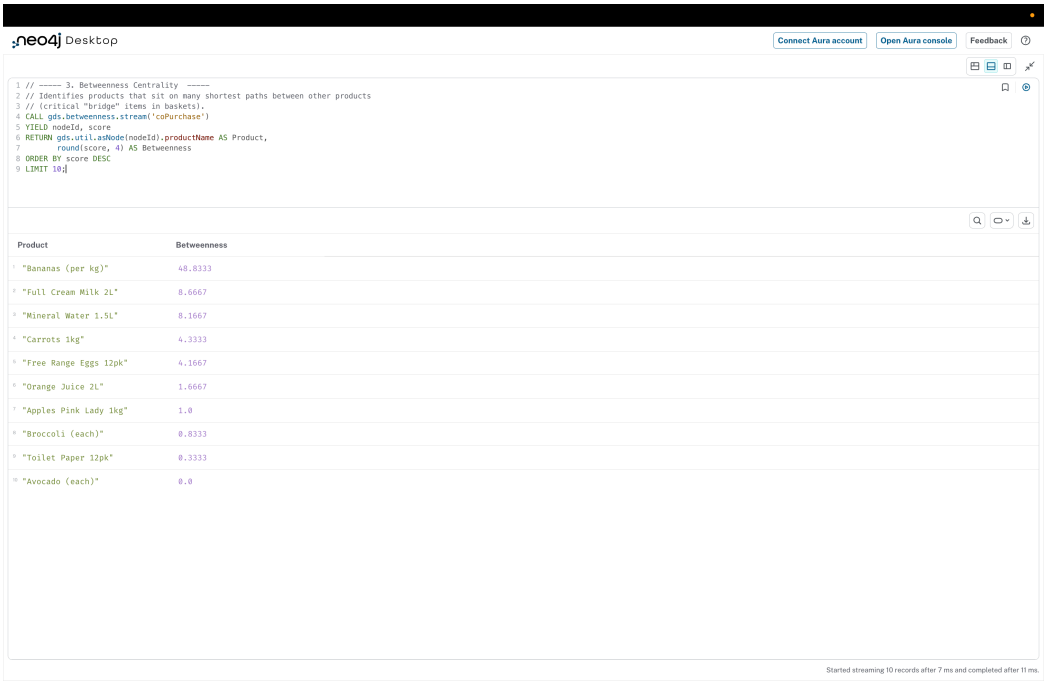


Figure 8 Betweenness Centrality results

5.4 Comparison and Recommendation

To evaluate the three centrality algorithms rigorously, rankings were compared using top-k overlap and qualitative interpretability for the retail use case. Table 10 summarises the top-5 products under each algorithm and the degree of agreement.

Table 10 Top-5 Product Rankings under PageRank, Degree Centrality and Betweenness Centrality

Rank	PageRank	Degree Centrality	Betweenness Centrality
1	Bananas (per kg) — 2.6273	Bananas (per kg) — 18.0	Bananas (per kg) — 48.83
2	Full Cream Milk — 1.9961	Full Cream Milk — 14.0	Full Cream Milk — 8.67
3	Mineral Water — 1.3422	Mineral Water — 8.0	Mineral Water — 8.17
4	Apples Pink Lady — 0.9721	Apples / Eggs / TP — 6.0	Carrots 1kg — 4.33
5	Free Range Eggs — 0.9297	(tied at 6.0)	Free Range Eggs — 4.17

Note: "Apples / Eggs / TP" are shown as a tied group at Degree Centrality 6.0, as provided.

Agreement analysis

Top-3 overlap is perfect: Bananas, Full Cream Milk and Mineral Water have a 100% top-3 overlap. The top 5 overlap is high: PageRank and Degree have same product set; Betweenness adds some of Degree's products with Carrots, a product that bridges between vegetable and staple clusters. It is therefore not surprising that PageRank is most closely correlated with Degree (practically the same ordering) and slightly less with Betweenness (which is optimized for a different objective (shortest-path brokerage) than PageRank (recursive prestige/connectivity)).

Algorithmic rationale and fitness for purpose

Recursive importance: a product that is bought in pairs with other higher scored products earns a higher score on PageRank. It's fitting when the question for your business is, which products increase basket value by association with other popular products? Simple baseline measures are also easy to calculate, and to explain to store managers, they are the raw numbers of weighted connections, known as Degree Centrality. Betweenness Centrality: which products are on a lot of shortest paths between otherwise independent basket communities; which products, if they didn't exist, would most break up typical shopping patterns?

Recommendation under evaluation criteria

When it comes to considering the risk of supply during the wet season, Degree Centrality is sufficient and most transparent for the store manager, with a slight refinement by PageRank by raising products that are linked to other hubs. Additionally, Betweenness provides complementary information for the assortment planning problem: Carrots have a higher score on Betweenness than Degree/PageRank: carrots are linked to purchases of

vegetables and staple foods. The team proposes thus two approaches: (1) the high Degree/PageRank items are available in the stores on a continuous basis (Bananas, Milk, Mineral Water); and (2) the inter-store transfers that are involved in the basket completeness should consider the high-Betweenness bridge items when designing.

Managerial recommendation: Bananas, Full Cream Milk and Mineral Water should be ensured to be available throughout the wet season if there are supply issues. A transfer guide is a degree/PageRank guide that is used to replenish, a Betweenness guide is used to assist assortments and transfer decisions of bridge items like Carrots.

6.1 Node Similarity – Jaccard

Top pairs:

Table 11 Top Jaccard pairs.

Product 1	Product 2	Jaccard
Apples Pink Lady 1kg	Avocado (each)	0.667
Carrots 1kg	Potatoes 2kg	0.500
Mineral Water 1.5L	Carrots 1kg	0.500
Free Range Eggs 12pk	Full Cream Milk 2L	0.429
Toilet Paper 12pk	Full Cream Milk 2L	0.429
Bananas (per kg)	Full Cream Milk 2L	0.333

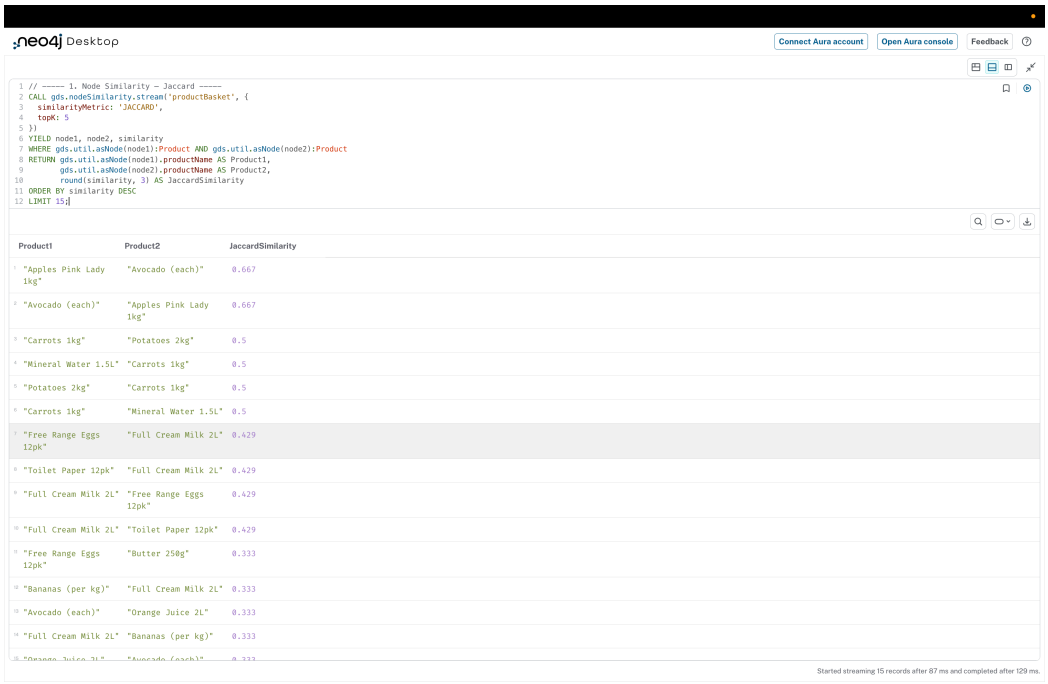


Figure 9 Node Similarity (Jaccard) product pairs

6.2 Node Similarity – Cosine

Top pairs:

Table 12 Top Cosine pairs.

Product 1	Product 2	Cosine
Apples Pink Lady 1kg	Avocado (each)	0.816
Carrots 1kg	Mineral Water 1.5L	0.707
Potatoes 2kg	Carrots 1kg	0.707
Toilet Paper 12pk	Full Cream Milk 2L	0.655
Free Range Eggs 12pk	Full Cream Milk 2L	0.655
Butter 250g	Free Range Eggs 12pk	0.577

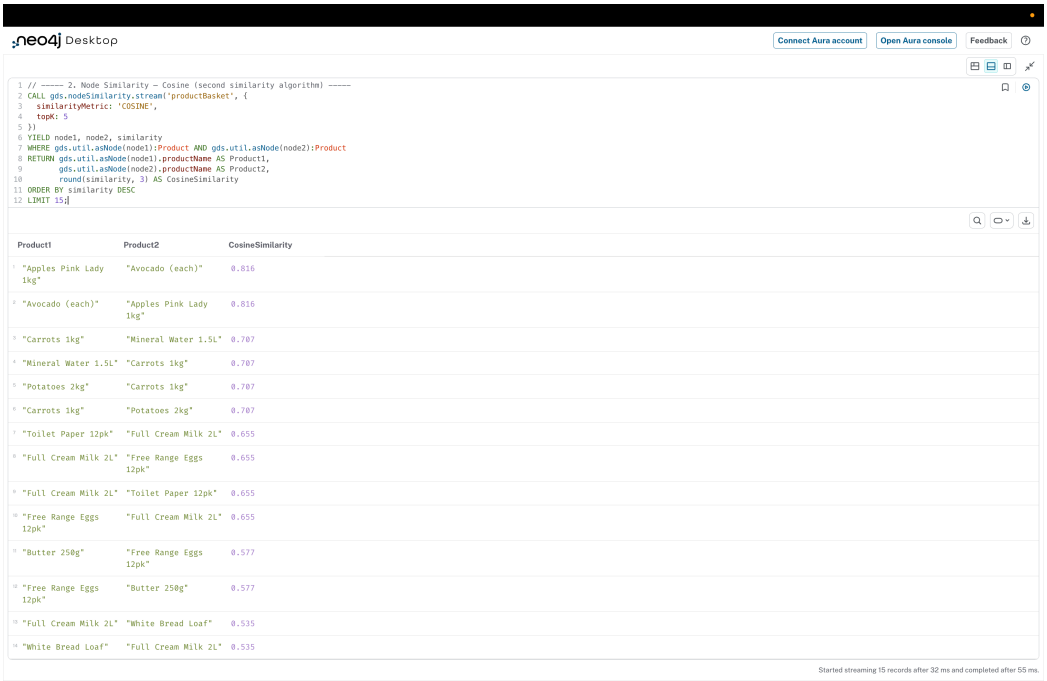


Figure 10 Node Similarity (Cosine) product pairs

6.3 Comparison and Recommendation

Jaccard and Cosine Node Similarity were evaluated on the same bipartite projection (Product–SalesTransaction via INCLUDES) with identical topK = 5. Table 13 compares the top pairs and score behaviour.

Table 13 Comparison of Top Product Pairs under Jaccard and Cosine Node Similarity

Pair	Jaccard	Cosine	Rank (J)	Rank (C)
Apples – Avocado	0.667	0.816	1	1
Carrots – Potatoes	0.500	0.707	2	3
Mineral Water – Carrots	0.500	0.707	3	2
Eggs – Full Cream Milk	0.429	0.655	4	5
Toilet Paper – Full Cream Milk	0.429	0.655	5	4

Metric behaviour and agreement

The rank correlation between Jaccard and Cosine is high, with only minor changes, where the same 5 pairs are ranked 1st, 2nd, 3rd, 4th and 5th. Cosine scores are generally higher than Jaccard scores, as Cosine scores normalise for the length of the vectors, which makes it less sensitive to the presence of a product in a higher number of transactions. Jaccard is more strict, evaluates pure overlap in sets and is likely to produce more conservative similarity estimates when the data is sparse in the form of baskets.

Fitness for the substitute-recommendation use case

Jaccard is better for “customers who bought X also bought Y” and “out-of-stock substitution” cases, where it is necessary to deliver highly accurate, conservative product suggestions (with fewer false positives). Cosine is better suited for situations where the coverage of the candidates matters and not their accuracy—such as creating more candidates for a promotions engine. The team recommends the use of Jaccard as the main substitute signal (Apples↔Avocado, Carrots↔Potatoes) and Cosine as a secondary expansion signal for Darwin multi-store applications in which there are sparse samples. Both of these metrics are easily recalculated after the daily POS load.

Cross-algorithm synthesis

Centrality and similarity addresses complementary questions. Centrality determines the SKUs that need to be carried to ensure the overall value of the basket (Bananas, Milk, Mineral Water). Similarity recognises which other SKUs are available to take sales when a hub or regional SKU is not available (Apples for Avocado and Potatoes for Carrots). Both families of algorithms can be used in concert: If the centrality of an item is high, the item is being monitored for replenishment priority; if the centrality of the item is not high, but the item is

comparable to a high-centrality item, it is being compared for replacing the item at the point of sale if the centrality of the item is below reorder level.

Managerial recommendation: If less than a reorder level SKU, use high similarity pairs (in particular Jaccard's conservative set) as basis for automatic suggestions to substitute in the POS system; If a wider candidate list needed, use Cosine.

7. Discussions and Limitations

The property-graph mapping retained the semantics of the EER for Assessment 2, while making changes to the multi-hop query ergonomics. The three centrality algorithms gave comparable rankings, which were consistent with company experience about the importance of well-known products, while the two-similarity metrics gave similar substitutes.

Limitations. The sample size is relatively small (22 products, 18 transactions), which reduces the power of the statistics. The co-purchase graph is a non-directed graph and it doesn't reflect sequence or seasonality. The units sold in the last 7 days were returned as zero due to a sample date being outside the rolling window. A pure-Cypher load path was used for Desktop compatibility. Due to the small sample size, the test for formal statistical significance was not used; larger POS extracts would enable bootstrap confidence intervals of PageRank and similarity scores, and a more conclusive study of the selection of algorithms.

Future work. Incorporate time-decayed relationships and larger POS extracts; apply Betweenness on the supplier–product network; experiment with FastRP + KNN; evaluate community-detection algorithms (Louvain, Leiden).

8. Conclusion

This assessment successfully migrated the Darwin multi-store supermarket relational design into a Neo4j property graph (86 nodes, 206 relationships), executed four Cypher use cases of increasing complexity, and demonstrated three centrality algorithms (PageRank, Degree, Betweenness) and two similarity algorithms (Jaccard and Cosine) with the Graph Data Science library. The results support practical decisions on promotion shortlists, loyalty engagement, low-stock transfers, supplier ranking, basket-hub prioritisation, and data-driven substitute recommendations.

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Individual Contributions

NiteshRajChaudhary(S394811) – Group Leader

Supervised overall coordination, quality control and final submission. Designed the Neo4j property-graph model (EER translation & optimisations). Implemented and interpreted all the three centrality algorithms (PageRank, Degree Centrality and Betweenness Centrality). Wrote the Centrality section of the report, along with comparative tables and management suggestions. Completed final report, included all figures, did final QC prior to submission.

Dilliram Dhakal (S396633)

Developed the graph visualisation (Neo4j Bloom) that was used as Figure 1. Wrote the Graph Database Modelling and Implementation sections, including reasoning about design decisions (inventories as relationships, specialisations as labels, weighted edges). Composed the support text for Discussions and Limitations section as well as the Conclusion.

Roshan Kumar Shrestha (S395498)

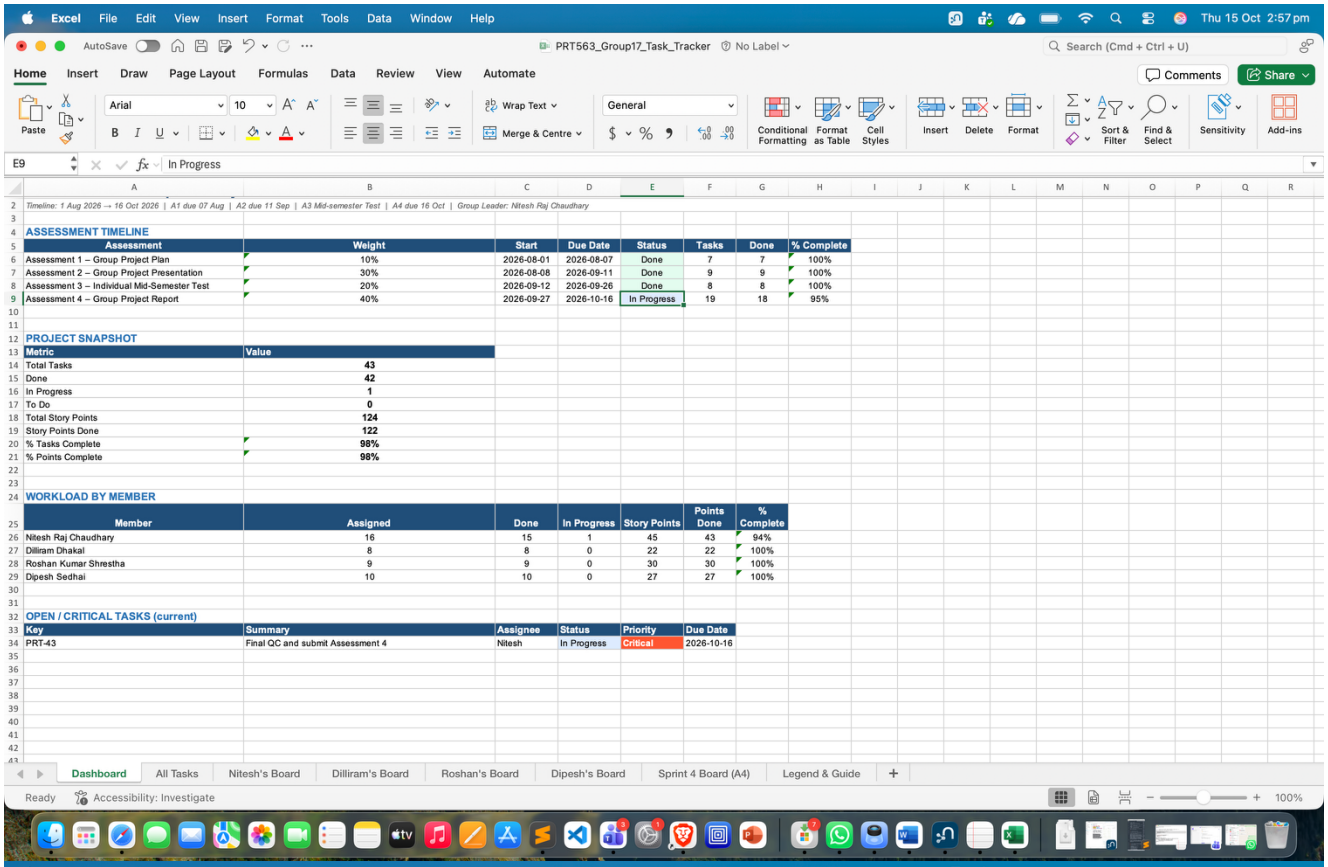
Installed all pure-Cypher import scripts (constraints, indexes, nodes, relationships and CO_PURCHASED materialisation). Created and quality-checked all the sample data files in CSV format. Wrote four Cypher query use cases of increasing complexity, ran them in Neo4j Desktop, and took screenshots of the results (Figures 2–5). Verified number of nodes and relationships after migration.

Dipesh Sedhai (S395457)

Adopted both similarity algorithms – Jaccard and Cosine Node Similarity with the help of Graph Data Science library. Screenshots all GDS algorithm results (Figures 6-9). Wrote the Similarity section of the report; provided metric comparison and recommendation of substitution. Compiled the repository (import.txt, queries.txt and CSV files) for submission and helped format the report and caption figures.

Appendix

Screenshots of team communication channel and task tracker.



ASSESSMENT TIMELINE

Assessment	Weight	Start	Due Date	Status	Tasks	Done	% Complete
Assessment 1 – Group Project Plan	10%	2026-08-01	2026-08-07	Done	7	7	100%
Assessment 2 – Group Project Presentation	30%	2026-08-08	2026-09-11	Done	9	9	100%
Assessment 3 – Individual Mid-Semester Test	20%	2026-09-12	2026-09-26	Done	8	8	100%
Assessment 4 – Group Project Report	40%	2026-09-27	2026-10-16	In Progress	19	18	95%

PROJECT SNAPSHOT

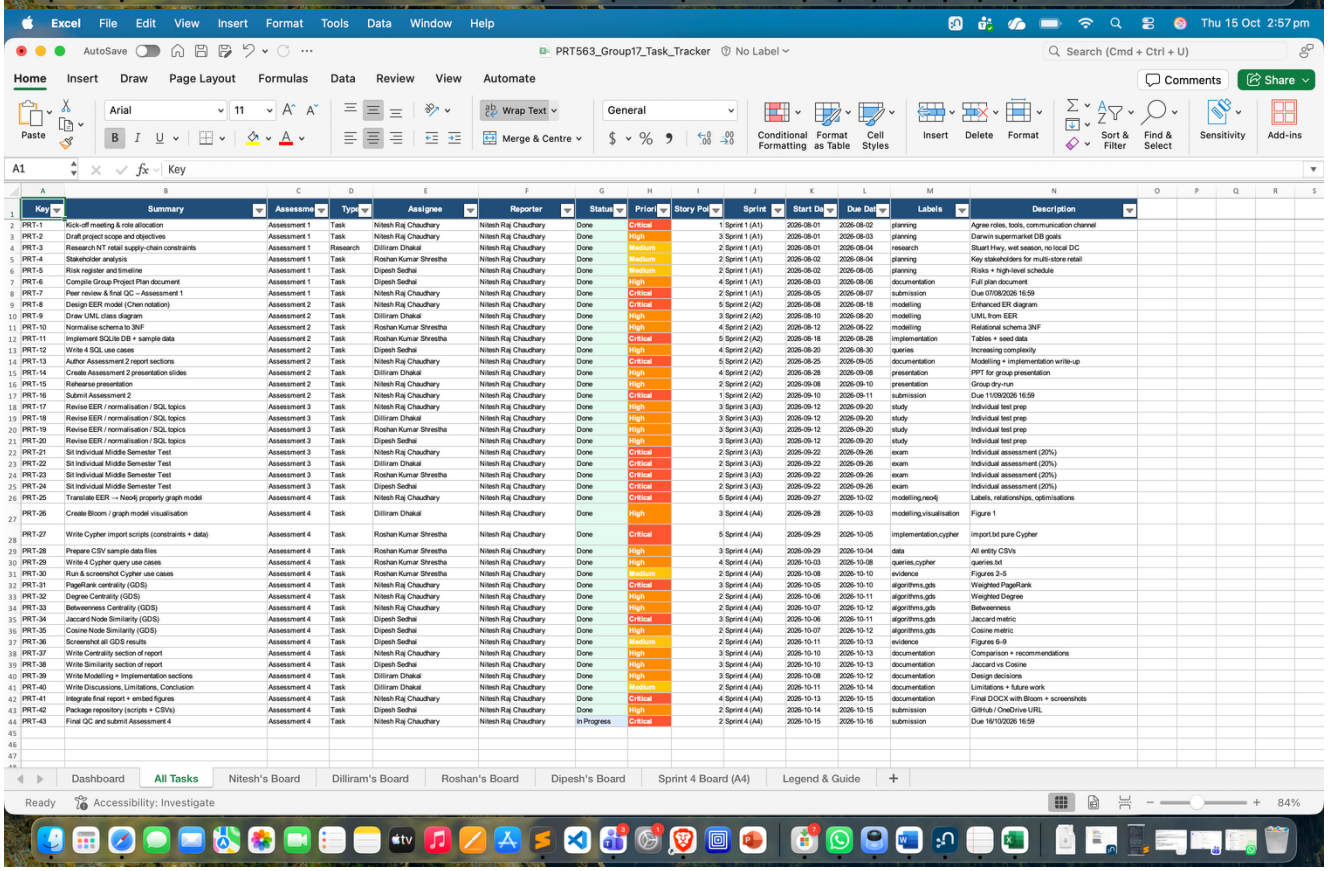
Metric	Value
Total Tasks	43
Done	42
In Progress	1
To Do	0
Total Story Points	124
Story Points Done	122
% Tasks Complete	98%
% Points Complete	98%

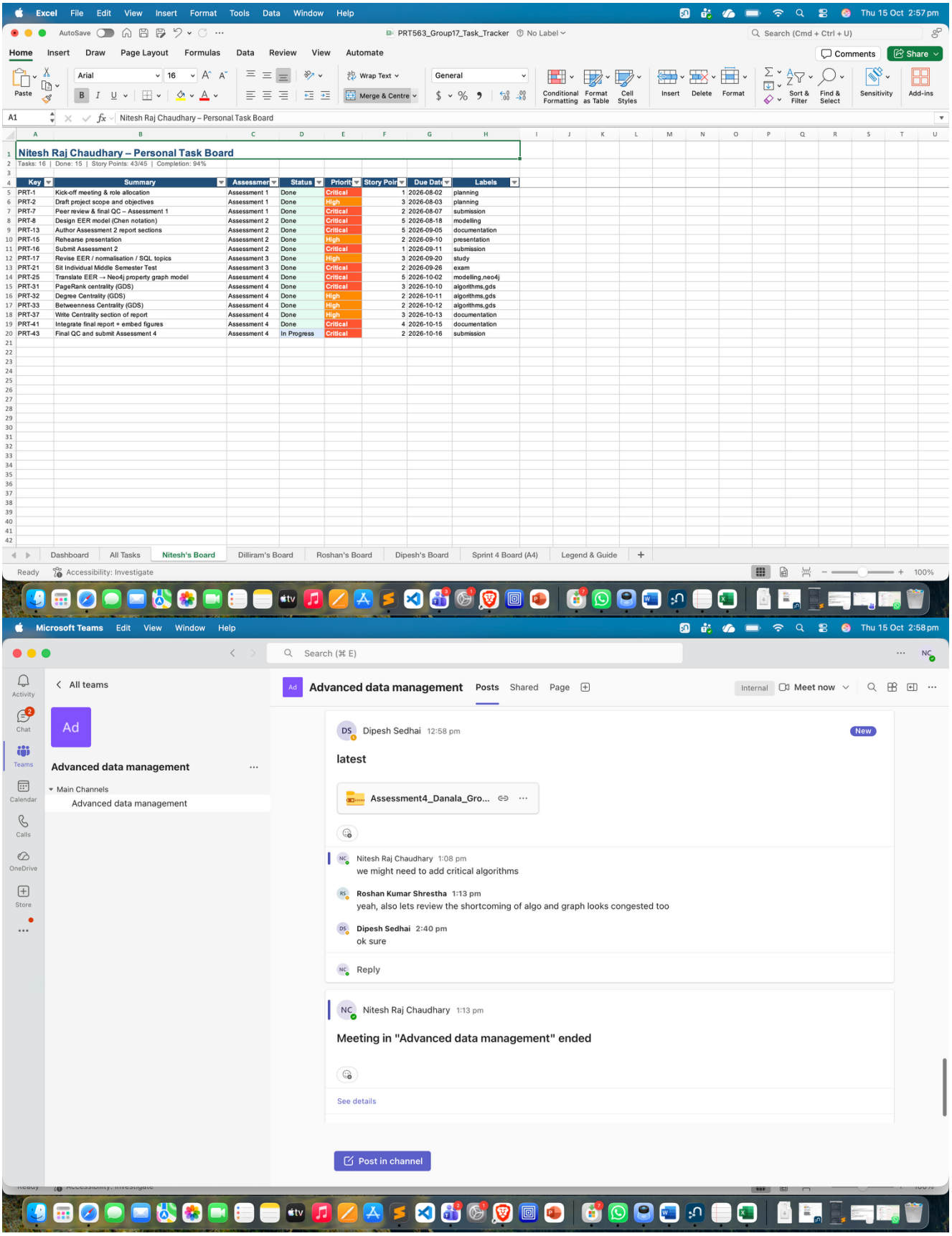
WORKLOAD BY MEMBER

Member	Assigned	Done	In Progress	Story Points	Points Done	% Complete
Nitesh Raj Chaudhary	16	15	1	45	43	94%
Dilliram Dhakal	8	8	0	22	22	100%
Roshan Kumar Shrestha	9	9	0	30	30	100%
Dipesh Sedhai	10	10	0	27	27	100%

OPEN / CRITICAL TASKS (current)

Key	Summary	Assignee	Status	Priority	Due Date
PRT-43	Final QC and submit Assessment 4	Nitesh	In Progress	Critical	2026-10-16





Nitesh Raj Chaudhary – Personal Task Board

Tasks: 16 | Done: 15 | Story Points: 43/45 | Completion: 94%

Key	Summary	Assessment	Status	Priority	Story Points	Due Date	Labels
PRT-1	Kick-off meeting & role allocation	Assessment 1	Done	Critical	1	2026-08-02	planning
PRT-2	Draft project scope and objectives	Assessment 1	Done	High	3	2026-08-03	planning
PRT-7	Peer review & final QC – Assessment 1	Assessment 1	Done	Critical	2	2026-08-07	submission
PRT-8	Design EER model (Chen notation)	Assessment 2	Done	Critical	5	2026-08-18	modelling
PRT-13	Author Assessment 2 report sections	Assessment 2	Done	Critical	5	2026-09-05	documentation
PRT-15	Rehearse presentation	Assessment 2	Done	High	2	2026-09-10	presentation
PRT-16	Submit Assessment 2	Assessment 2	Done	Critical	1	2026-09-11	submission
PRT-17	Revise EER / normalisation / SQL topics	Assessment 3	Done	High	3	2026-09-20	study
PRT-21	Sit Individual Middle Semester Test	Assessment 3	Done	Critical	2	2026-09-26	exam
PRT-25	Translate EER → Neo4j property graph model	Assessment 4	Done	Critical	5	2026-10-02	modelling,neo4j
PRT-31	PageRank centrality (GDS)	Assessment 4	Done	Critical	3	2026-10-10	algorithms,gds
PRT-32	Degree Centrality (GDS)	Assessment 4	Done	High	2	2026-10-11	algorithms,gds
PRT-33	Betweenness Centrality (GDS)	Assessment 4	Done	High	2	2026-10-12	algorithms,gds
PRT-37	Write Centrality section of report	Assessment 4	Done	High	3	2026-10-13	documentation
PRT-41	Integrate final report + embed figures	Assessment 4	Done	Critical	4	2026-10-15	documentation
PRT-43	Final QC and submit Assessment 4	Assessment 4	In Progress	Critical	2	2026-10-16	submission

Dashboard | All Tasks | Nitesh's Board | Dilliram's Board | Roshan's Board | Dipesh's Board | Sprint 4 Board (A4) | Legend & Guide

Advanced data management

Posts | Shared | Page

Dipesh Sedhai 12:58 pm — New

latest

Assessment4_Danala_Gro...

Nitesh Raj Chaudhary 1:08 pm
we might need to add critical algorithms

Roshan Kumar Shrestha 1:13 pm
yeah, also lets review the shortcoming of algo and graph looks congested too

Dipesh Sedhai 2:40 pm
ok sure

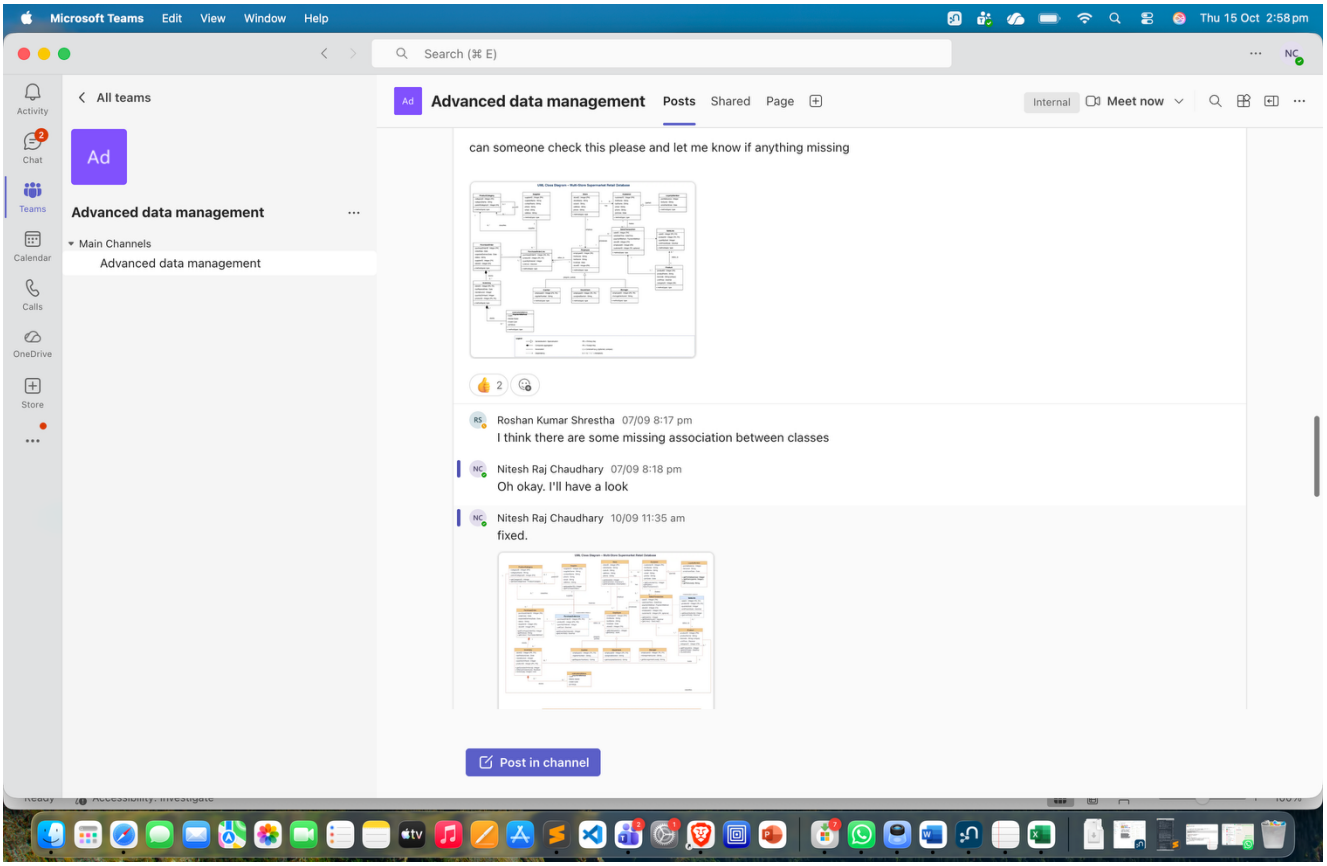
Reply

Nitesh Raj Chaudhary 1:13 pm

Meeting in "Advanced data management" ended

See details

Post in channel


Microsoft Teams Edit View Window Help
Thu 15 Oct 2:58 pm
Search (⌘ E)
All teams
Ad
Advanced data management
Main Channels
Advanced data management
Advanced data management
Posts
Shared
Page
Internal
Meet now
can someone check this please and let me know if anything missing
Roshan Kumar Shrestha 07/09 8:17 pm
I think there are some missing association between classes
Nitesh Raj Chaudhary 07/09 8:18 pm
Oh okay. I'll have a look
Nitesh Raj Chaudhary 10/09 11:35 am
fixed.
Post in channel